

WHERE IT STARTED

Extremes, not averages

My first research work was at the Institute of Applied Physics of the Academy of Sciences, processing signal data on extreme ocean wave phenomena. Most of a wave record is unremarkable — it's the rare, oversized event that determines whether a structure survives. That project set the direction for everything after it: I am consistently more interested in the *tail of the distribution* than in the mean. Extreme value theory, anomaly detection, rare-event statistics — this isn't a subject I picked for career reasons, it's the question I keep finding on the other side of every dataset.

HOW I ACTUALLY THINK

Where the model breaks

A model is easy to build and easy to trust — right up until it meets data it wasn't shaped for. I don't consider a model finished when it fits; I consider it finished when I've found the point where its assumptions stop holding, proven *why* they stop holding, and decided what to do about it. That's the part of the work most people skip, and it's the part I find genuinely interesting. It's also, not coincidentally, where the real risk in any deployed system lives.

OUTSIDE THE EQUATIONS

What I read the same way

I apply the same instinct outside of math. I watch horror cinema less for the scare and more for the *mechanism* — the specific structural choices a director uses to build dread before anything happens on screen. I read around neuroscience and philosophy of mind for the same reason I read

extreme value theory: both are attempts to find the load-bearing structure underneath something that looks, on the surface, like noise. And I follow macroeconomics and political-economic systems the way I'd audit a model — tracing where the official narrative and the underlying structure diverge, and by how much.

HOW I WORK

The complete cycle

I don't stop at the proof or the notebook. `116117-monitor.tech` — a FastAPI and Playwright service I built and still operate on my own DigitalOcean infrastructure, monitoring German medical appointment availability in real time — exists because I wanted the full loop: theorem to pipeline to server to something a real user can open in a browser. That's the differentiator I'd want a reader of this page to take away: *the complete cycle, from complex math to a deployed container*. I'm equally comfortable proving a bound and administering the Nginx config that serves the result.

WHERE THINGS STAND

No shortcuts taken

I finished my MSc in Computational Mathematics at the Universität Passau in March 2026, on top of two Springer/Scopus publications, a proven theorem, a commercialized ML model, a personal research grant, and nineteen international conference presentations — built while working delivery shifts to cover the gap the job market left open. I mention that not for sympathy but because it's accurate context: nothing here was handed to me, and I don't expect the next thing to be either.